Notes 3-12

Using our 3-10 notes, using the quotient remainder thm,

the $\gcd(10, a) = \gcd(a, r)$

$= \gcd(r, r_1)$

$= \gcd(r_1, r_2)$

$\ldots$

Suffices to prove

d is a divisor of a & b if and only if

d is a divisor of r and a.

I.e.

$\{$common divisors of a and $b\} = \{$common divisors of a and $r\}$

Suppose $d|a$ and $d|b$. By def of divides, $a = d \cdot c$

and $b = d \cdot e$, for some $c, e \in \mathbb{Z}$.

Thus, we have $b = a \cdot q + r$, can be written as $d \cdot e = d \cdot c \cdot q + r$.

Hence, $d \cdot e - d \cdot c \cdot q = r$ so $d(e - cq) = r$. Therefore, $d | r$.

Conversely suppose $w | a$ and $w | r$. Thus $a = w \cdot s$ and $r = w \cdot t$.

Thus $\exists\, s, e \in \mathbb{Z}$ such that $a = ws$ and $r = wt$. We see that

$b = aq + r = wsq + wt = w(sq + t)$. Hence $w | b$. $\square$

We conclude that since $d|a$ and $d|b \Leftrightarrow d|a$ and $d|r$,

$\gcd(a, b) = \gcd(a, r)$. $\square$

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